

Topic Wise Practice Assignment

Python Basics- Practice assignment

1 Write a python program to Print following statements as an output using print statement.

Student Name:
Address:
Contact _No:
Mother Tongue:
School _Name:
Year:
Panel:
Roll_No:

2 In the previous code you written, modify the statements printing following fields into multi-line comments, so these fields will not be the part of the output.

Address:
Contact _No:
Mother Tongue

3 Accept Student Name, Roll Number and Marks of the 3 subjects from the user. Calculate the percentage of the marks and display it. Display the Subject with Highest and lowest marks.

4 write a python code to check given number is positive, negative or zero

5 write a python code to check given number is even or odd

6 Given two non-negative values, print true if they have the same last digit such as 27 and 57.

7 Write a python program to print numbers from 1 to 10 in single row with one tab space

8 Write a python program to print even numbers between 23 to 57. Each number should be printed in a separate row.

9 Write a python program to check if given number prime or not

10 Write a python program to print prime numbers between 10 to 99

11 Write a python program to print sum of all the digits of given number

12 Write a program to reverse a given number and print

13 Write a program to find if the given number is palindrome or not

14 Accept 5 numbers from the user and display their cube values.

15 Display prime factors of number. Accept number from user.

16 Display following pattern

```
*  
* *
```

```

* * *
* * * *

```

17 Display following pattern

```

      *
    * *
  * * *
* * * *

```

Mini Project

1: Create a python program that asks the user how far they want to travel. If they want to travel less than 3 miles tell them to ride a bicycle, if they want to travel more than 3 miles but less than 300 miles tell them to ride a motorcycle. If they want to travel 300 miles or more tell them to driver supercar

2: Lets assume you are planning to use your python skills to build PBLApp for mobile. You decide to host your application on servers running in the cloud. You pick a hosting provider that charges \$0.51 per hour. You will launch your service using one server and want to know how much it will cost to operate per day, per week, per month

Write a python code that displays answers of following questions

How much does it costs to operate one server per day?

How much does it costs to operate one server per week?

How much does it costs to operate one server per month?

How many days can I operate one server with \$918?

Data Structures- Practice assignment

1)List

- 1 Write a program to create a list of 5 integers and display the list items. Access individual elements through index.
- 2 Write a program to append a new item to the end of the list.
- 3 Write a program to reverse the order of the items in the list.
- 4 Write a program to print the number of occurrences of a specified element in a list.
- 5 Write a program to append the items of list1 to list2 in the front.
- 6 Write a program to insert a new item before the second element in an existing list.
- 7 Write a program to remove the item from a specified index in a list.
- 8 Write a program to remove the first occurrence of a specified element from a list.
- 9 Accept 20 values from user and save it in list. Perform following operations on it
 - a) count similar elements of list and print their index value
 - b) count even and odd values of list
 - c) count positive and negative values of list.
- 10 Accept 10 values from user and save it in list. Perform following operations on it

- a) Sort list in ascending order using sorted() function and display sorted list
 - b) sort list in descending order using sort() function
 - c) display length of list
- 11 Accept two lists from user and merge them using + in a single list
- 12 An acronym is a word formed by taking the first letters of the words in a phrase and making a word from them. For example, RAM is an acronym for “random access memory.” Write a program that allows the user to type in a phrase and then outputs the acronym for that phrase. Note: the acronym should be all uppercase, even if the words in the phrase are not capitalized.
- 13 Write a program to print the abbreviation of a month, given its number

2) Dictionary

- 1 Write a program to add a key and value to a dictionary.
 Sample Dictionary: (0:10, 1: 20)
 Expected Result: (0: 10, 1:20, 2:30)
- 2 Write a program to concatenate the following dictionaries to create a new one.
 Sample Dictionary:
 dic1=(1:10, 2:20) dic2={3:30, 4:40} dic3={5:50,6:60}
 Expected Result (1: 10, 2: 20, 3: 30, 4: 40, 5:50, 6: 60)
- 3 Write a program to check if a given key already exists in a dictionary.
- 4 Write a program to iterate over a dictionary using for loop and print the keys alone, values alone and both keys and values.
- 5 Write a program to prepare a dictionary where the keys are numbers between 1 and 15 (both included) and the values are square of the keys.
- 6 Write a program to sum all the values in a dictionary, considering the values will be of int type.
- 7 Assign 5 elements in dictionary named my_dict. This dictionary contains student's information. Add and delete student information from my_dict. Display student information
- 8 Convert list to dictionary
 e.g. list1=['name', 'panel', 'rollno']
 list2=['ABC', 'B', 34]
 my_dict= { 'name': 'ABC', 'panel': 'B', 'rollno': 34 }
- 9 Convert dictionary to lists. Make one list of keys and other of values
- 10 Find the mean of values of dictionary
 e.g mydict= { 'marks1': 23, 'marks2': 123, 'marks3': 43, 'marks4': 13, 'marks5': 39)
 output: mean of all values.
- 11 Sort the elements of the dictionary using sorted() function

```
my_dict= { 'name': 'ABC', 'panel'='B', 'rollno'=34, 'marks'=[65,87,67,94]}  
sorted(my_dict)
```

3) Tuple

- 1 Write a program to print the 4th element from first and 4th element from last in a tuple.
- 2 Write a program to check whether an element exists in a tuple or not.
- 3 Write a program to convert a list into a tuple.
- 4 Write a program to find the index of an item in a tuple.
- 5 Write a program to replace last value of tuples in a list to 100.

Sample list: [(10, 20, 40), (40, 50, 60), (70, 80, 90)]

Expected Output: [(10, 20, 100), (40, 50, 100), (70, 80, 100)]

4) Set

- 1 Write a program to remove a given item from the set.
- 2 Write a program to create an intersection of sets.
- 3 Write a program to create an union of sets.
- 4 Write a program to find the maximum and minimum value in a set.

5) String

- 1 Write a program to count the number of upper and lower case letters in a String.
- 2 Write a program that will check whether a given String is Palindrome or not.
- 3 Given a string, return a new string made of n copies of the first 2 chars of the original string where n is the length of the string. The string length will be ≥ 2 .
If input is "Wipro" then output should be "Wiwiwiwi".
- 4 Given a string, if the first or last character is 'x', return the string without those 'x' character, else return the string unchanged. If the input is "xHix", then output is "Hi".
- 5 Given a string and an integer n, return a string made of n repetitions of the last n characters of the string. You may assume that n is between 0 and the length of the string inclusive. For example if the inputs are "Wipro" and 3, then the output should be "propopro".

Regular Expression

Create a Regular Expression and implement the following

- a) Recognize the following strings: "bat," "bit," "but," "hat," "hit," or "hut."
- b) Match any pair of words separated by a single space, i.e., first and last names.
- c) Match any word and single letter separated by a comma and single space, as in last

name, first initial.

6.Numpy

1. Create a 3×3 NumPy array of all True
2. Create a 1D array of 10 evenly spaced values between 5 and 50
3. Convert a Python list into a NumPy array
4. Reverse a 1D NumPy array
5. Create a 3×3 identity matrix
6. Create a 4×4 array and extract its first row and last column
7. Write a code to Create a 4×4 NumPy array and extract the first two rows and first two columns.
8. Perform arithmetic operations on two NumPy arrays element-wise
9. Write a code to compute the dot product of two NumPy arrays
10. Compute the mean, median, and standard deviation of a NumPy array